

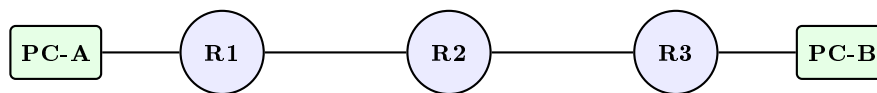
HIMALAYA COLLEGE OF ENGINEERING

Department of Computer Engineering
(Affiliated to Tribhuvan University / Purbanchal University)

LABORATORY MANUAL

Static and Dynamic Routing

RIP · OSPF · BGP – Cisco Packet Tracer



Course: Computer Networks / Data Communication

Programme: B.E. Computer Engineering

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Tool required: **Cisco Packet Tracer 8.x** (or later)
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Introduction

This manual contains four hands-on laboratory exercises that build a progressively deeper understanding of IP routing using **Cisco Packet Tracer**. Students move from manually configured **static routes** to the three dynamic routing protocols most commonly taught in an undergraduate networking course: **RIP (distance-vector)**, **OSPF (link-state)**, and **BGP (path-vector, inter-domain)**.

General Instructions for All Labs

- Build the topology exactly as shown before entering any configuration commands.
- Use **Copper Straight-Through** cables between a router/switch and an end device, and **Copper Straight-Through** (auto-MDIX) between two routers' FastEthernet/GigabitEthernet interfaces. Serial links use **Serial DCE/DTE** cables (the DCE end must have **clock rate** configured).
- Enter Privileged EXEC mode with **enable**, then Global Configuration mode with **configure terminal** before applying any of the configuration steps below.
- After completing each lab, save the configuration with **copy running-config startup-config** (or **write memory**).
- Verify every lab using the **show** commands listed in the *Verification* subsection, and take a screenshot of the successful **ping/traceroute** for your lab report.

Static vs. Dynamic Routing at a Glance

Aspect	Static Routing	Dynamic Routing
Configuration	Manually entered by the administrator	Automatically learned via a routing protocol
Scalability	Poor for large/changing networks	Good – adapts to topology changes
CPU/Bandwidth overhead	None	Periodic or triggered updates consume resources
Convergence	Instant (fixed), but no re-routing on failure	Re-converges automatically after a link/router failure
Typical use	Small networks, stub networks, default routes	Medium-large networks, enterprise/ISP backbones

Protocol	Type	Metric	Default Administrative Distance
Static route	–	–	1
RIP (v2)	Distance-vector (IGP)	Hop count (max 15)	120
OSPF	Link-state (IGP)	Cost (10^8 /bandwidth)	110
eBGP	Path-vector (EGP)	AS-path / attributes	20
iBGP	Path-vector (EGP)	AS-path / attributes	200

LAB 1: Static Routing

Learning Objectives

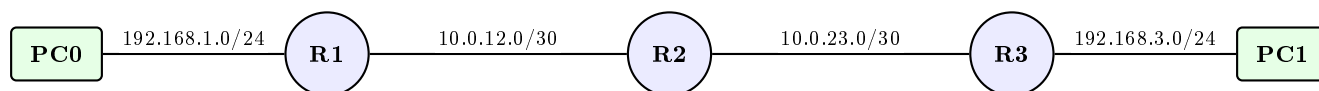
- Understand how static routes are manually configured on a Cisco router.
- Configure IP addressing on router and PC interfaces in Packet Tracer.
- Configure static routes (including a default/gateway-of-last-resort route) to achieve full end-to-end connectivity across a three-router network.
- Verify routing behaviour using `show ip route`, `ping`, and `tracert`.

1.1 Theory

A **static route** is a manually configured path from a router to a specific destination network. It is entered by the network administrator using the global configuration command `ip route`. Static routes do not adapt automatically to topology changes – if a link fails, traffic using that static route is dropped until the administrator intervenes (unless a floating static route with a higher administrative distance is configured as backup). Static routing is preferred for small, stub, or tightly controlled networks because it consumes no CPU or bandwidth for route computation and is fully predictable.

1.2 Topology

Three routers connect three separate LANs in series (R1 – R2 – R3), each router also serving a local PC subnet.



IP Addressing Table

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.10	255.255.255.0
R1	Gig0/0	192.168.1.1	255.255.255.0
R1	Gig0/1	10.0.12.1	255.255.255.252
R2	Gig0/0	10.0.12.2	255.255.255.252
R2	Gig0/1	10.0.23.1	255.255.255.252
R3	Gig0/0	10.0.23.2	255.255.255.252
R3	Gig0/1	192.168.3.1	255.255.255.0
PC1	FastEthernet0	192.168.3.10	255.255.255.0

1.3 Requirements

3 × Router (e.g. 1941 or 2911), 2 × PC, appropriate copper straight-through cables, Cisco Packet Tracer 8.x.

Procedure

Step 1 – Interface addressing (all routers).

```
Router> enable
```

```
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 10.0.12.1 255.255.255.252
R1(config-if)# no shutdown
```

Repeat the equivalent addressing on R2 and R3 using the table above. Also configure PC0 and PC1 with their IP address, subnet mask, and default gateway (192.168.1.1 for PC0, 192.168.3.1 for PC1) via *Desktop* → *IP Configuration*.

Step 2 – Static routes on R1. R1 only needs a route toward the far networks; a single default route is simplest.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 10.0.12.2
```

Step 3 – Static routes on R2 (transit router, both directions required).

```
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.12.1
R2(config)# ip route 192.168.3.0 255.255.255.0 10.0.23.2
```

Step 4 – Static route on R3.

```
R3(config)# ip route 0.0.0.0 0.0.0.0 10.0.23.1
```

Step 5 – Save configuration.

```
R1# copy running-config startup-config
```

(repeat on R2 and R3.)

Note

Using 0.0.0.0 0.0.0.0 as the destination/mask on the edge routers (R1, R3) creates a *default route* (gateway of last resort) so you do not need to list every remote network individually. On the transit router (R2), however, both remote networks must be listed explicitly because R2 sits between two stub networks with no single "default" direction.

1.4 Verification

```
R1# show ip route
R2# show ip route static
PC0> ping 192.168.3.10
PC0> tracert 192.168.3.10
```

A successful lab shows all three routers with the expected static/connected routes, and PC0 can ping PC1 with 4/4 successful replies.

1.5 Review Questions

1. What administrative distance does a static route have by default, and why does this matter when both a static and a dynamic route exist for the same destination?

2. What is a *floating static route* and when would you use one?
3. Why was a default route sufficient on R1 and R3 but not on R2?

Lab Exercise

Add a fourth router R4 with its own LAN (192.168.4.0/24) attached to R3. Reconfigure the static routes on all four routers (including changing R3 from a default route to explicit routes, since R3 is now also a transit router) so that every PC can reach every other PC.

LAB 2: Dynamic Routing with RIP (RIPv2)

Learning Objectives

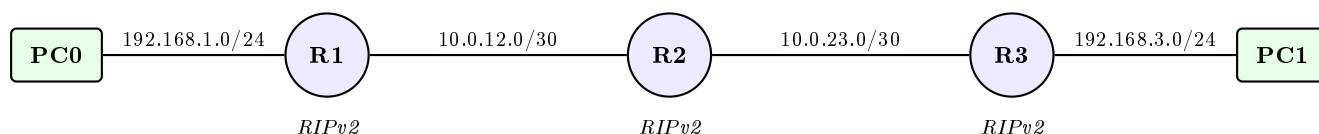
- Understand distance-vector routing and the Routing Information Protocol (RIP).
- Configure RIP version 2 on multiple routers in Packet Tracer.
- Observe automatic route propagation and disable auto-summarization.
- Verify RIP-learned routes and neighbor relationships.

2.1 Theory

RIP is a classic *distance-vector* Interior Gateway Protocol (IGP) that selects paths based purely on **hop count**, with a maximum usable value of 15 (16 = unreachable). Routers running RIP broadcast/multicast their entire routing table to directly connected neighbors every 30 seconds (periodic update). RIPv2 improves on RIPv1 by supporting **classless routing (VLSM)**, multicast updates (224.0.0.9), and simple authentication. Because it only considers hop count, RIP can pick a slower path over a faster one if the slower path has fewer hops – one of its major limitations, along with slow convergence and the 15-hop network diameter limit.

2.2 Topology

Same physical three-router topology as Lab 1, but routing is now automatic.



Use the same IP addressing table as Lab 1 (Section 1).

Procedure

Step 1 – Address the interfaces exactly as in Lab 1, Step 1 (or reuse the same Packet Tracer file and simply remove any static routes with `no ip route ...`).

Step 2 – Enable RIPv2 on R1.

```
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# no auto-summary
R1(config-router)# network 192.168.1.0
R1(config-router)# network 10.0.12.0
```

Step 3 – Enable RIPv2 on R2.

```
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 10.0.12.0
R2(config-router)# network 10.0.23.0
```

Step 4 – Enable RIPv2 on R3.

```
R3(config)# router rip
```

```
R3(config-router)# version 2
R3(config-router)# no auto-summary
R3(config-router)# network 10.0.23.0
R3(config-router)# network 192.168.3.0
```

Note

The **network** command under **router rip** does *not* take a subnet mask; it takes the **classful major network** (or the directly connected subnet in classless mode) and simply tells the router which of its own interfaces to run RIP on and advertise. Always issue **no auto-summary** in RIPv2 labs with VLSM/discontiguous subnets, otherwise routes may be summarized at classful boundaries and connectivity will break.

2.3 Verification

```
R2# show ip protocols
R2# show ip route rip
R1# show ip route
R1# debug ip rip
PC0> ping 192.168.3.10
```

In **show ip route**, RIP-learned routes are marked with the code R. Use **debug ip rip** (then **undebug all** to stop) to watch periodic update exchanges live in the simulation.

2.4 Review Questions

1. Why might RIP choose a 3-hop path over a 1-hop satellite link with much lower bandwidth?
2. What problem does *split horizon* solve in distance-vector protocols, and is it enabled by default in Packet Tracer's RIP implementation?
3. Compare the administrative distance of RIP (120) with static (1) and OSPF (110). If all three had a route to the same destination, which would be installed in the routing table?

Lab Exercise

Add a fourth router with a directly connected LAN and enable RIPv2 on it. Use **show ip route** on the two end routers to confirm the new network is learned automatically – no manual route entry required. Then simulate a link failure (turn off an interface) and observe RIP's convergence time using the Packet Tracer clock.

LAB 3: Dynamic Routing with OSPF (Single Area)

Learning Objectives

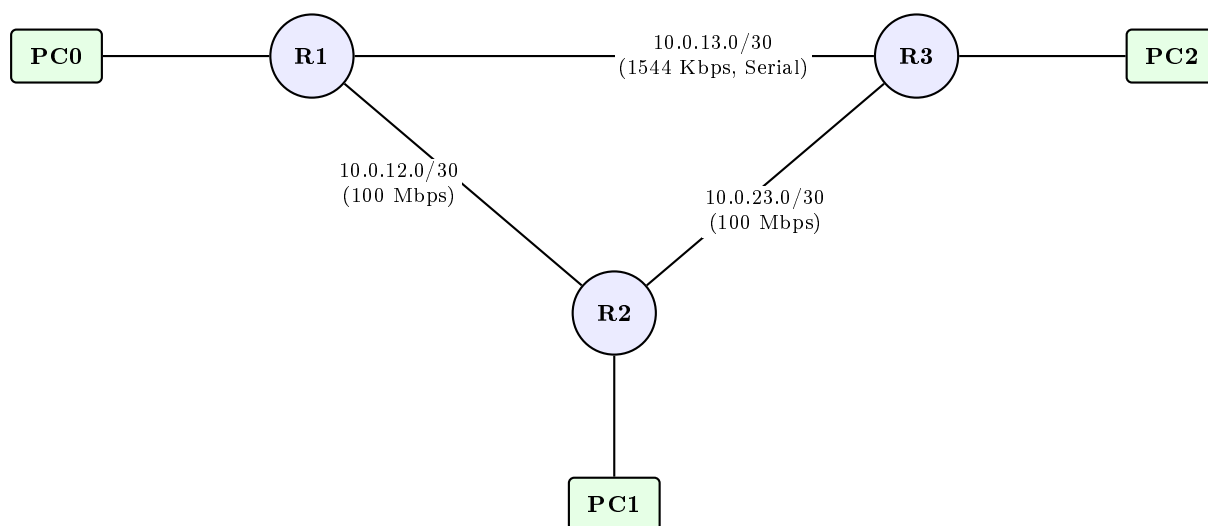
- Understand link-state routing concepts: LSAs, the SPF (Dijkstra) algorithm, and cost-based path selection.
- Configure single-area OSPF (Area 0) on multiple routers.
- Tune interface cost/bandwidth and observe the effect on route selection.
- Verify OSPF neighbor adjacencies and the link-state database.

3.1 Theory

OSPF (Open Shortest Path First) is a *link-state* IGP. Every router floods information about its directly connected links (Link-State Advertisements, LSAs) to all routers in the same area, so that every router builds an identical **link-state database (LSDB)** and independently computes the shortest path tree using **Dijkstra's SPF algorithm**. Path selection is based on **cumulative cost** ($\text{cost} = 10^8 / \text{bandwidth}(\text{bps})$ by default), which allows OSPF to correctly prefer high-bandwidth links over low-bandwidth ones – unlike RIP. OSPF routers form neighbor **adjacencies** using Hello packets (default 10 s on broadcast links) and elect a **Designated Router (DR)** and **Backup DR (BDR)** on multi-access segments to reduce flooding overhead.

3.2 Topology

A triangular (partial-mesh) topology is used so that OSPF's cost-based best-path selection can be clearly observed, in addition to a stub LAN on each edge router.



IP Addressing Table

Device	Interface	IP Address	Mask
R1	Gig0/1 (to PC0 LAN)	192.168.1.1	/24
R1	Gig0/0 (to R2)	10.0.12.1	/30
R1	Serial0/0/0 (to R3)	10.0.13.1	/30
R2	Gig0/1 (to PC1 LAN)	192.168.2.1	/24
R2	Gig0/0 (to R1)	10.0.12.2	/30
R2	Gig0/2 (to R3)	10.0.23.1	/30
R3	Gig0/1 (to PC2 LAN)	192.168.3.1	/24
R3	Gig0/2 (to R2)	10.0.23.2	/30
R3	Serial0/0/0 (to R1)	10.0.13.2	/30

All routers use **Process ID 1** and reside in **Area 0** (single-area/backbone OSPF, the simplest and most common configuration in an introductory lab).

Procedure

Step 1 – Address all interfaces per the table above (and configure PC0/PC1/PC2 with their gateway).

Step 2 – Set serial clock rate on the DCE end (required only on the side holding the DCE cable end, check with `show controllers serial0/0/0`):

```
R1(config)# interface serial0/0/0
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
```

Step 3 – Enable OSPF on R1.

```
R1(config)# router ospf 1
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.0.12.0 0.0.0.3 area 0
R1(config-router)# network 10.0.13.0 0.0.0.3 area 0
```

Step 4 – Enable OSPF on R2.

```
R2(config)# router ospf 1
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# network 10.0.12.0 0.0.0.3 area 0
R2(config-router)# network 10.0.23.0 0.0.0.3 area 0
```

Step 5 – Enable OSPF on R3.

```
R3(config)# router ospf 1
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
R3(config-router)# network 10.0.23.0 0.0.0.3 area 0
R3(config-router)# network 10.0.13.0 0.0.0.3 area 0
```

Step 6 (optional) – Router IDs and passive interfaces.

```
R1(config-router)# router-id 1.1.1.1
R1(config-router)# passive-interface gigabitEthernet 0/1
```

Setting an explicit router ID avoids ambiguity, and marking the LAN-facing interface passive stops unnecessary Hello packets from being sent toward end-user PCs while still advertising that LAN's network.

Note

OSPF's **network** command uses a **wildcard mask** (inverse of the subnet mask), not a normal subnet mask – e.g. 0.0.0.255 for a /24 and 0.0.0.3 for a /30. This is a very common point of confusion for students transitioning from RIP/static routing syntax.

3.3 Verification

```
R1# show ip ospf neighbor
R1# show ip route ospf
R1# show ip protocols
R1# show ip ospf interface brief
PC0> tracert 192.168.3.10
```

Confirm all three routers show each other as **FULL** state neighbors, and that the route from PC0 to PC2's LAN takes the R1–R2–R3 path (100 Mbps links, lower total cost) *rather than* the direct but slow 1.544 Mbps serial link R1–R3, even though the serial link is a shorter (fewer-hop) path – this demonstrates OSPF's cost-based (not hop-based) decision making, in contrast to RIP.

3.4 Review Questions

1. Explain why OSPF chose the two-hop Gigabit path over the one-hop but slow serial link, and contrast this with how RIP would have behaved on the same topology.
2. What is the purpose of the DR/BDR election, and on which link types does it occur?
3. What OSPF state indicates a fully synchronized adjacency, and what are the states that precede it?

Lab Exercise

Change the bandwidth of the R1–R3 serial link (**bandwidth** command, kbps) to a very high value and observe whether OSPF's chosen best path changes. Explain your observation using the OSPF cost formula $10^8/\text{bandwidth}$.

LAB 4: Inter-Domain Routing with BGP

Learning Objectives

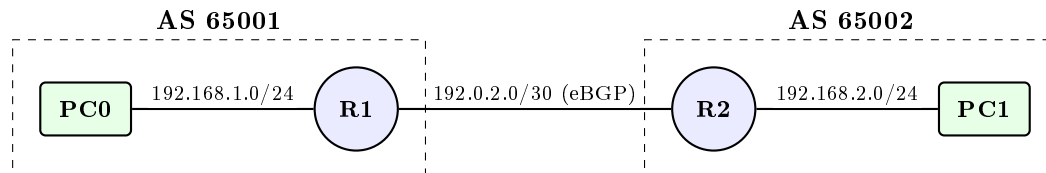
- Understand the role of BGP as the Exterior Gateway Protocol (EGP) of the Internet.
- Differentiate between eBGP (between autonomous systems) and iBGP (within an AS).
- Configure eBGP peering between two autonomous systems in Packet Tracer.
- Advertise networks into BGP and verify the BGP table and IP routing table.

4.1 Theory

BGP (Border Gateway Protocol) is the *path-vector* protocol that glues together the independently administered networks (**Autonomous Systems**, ASes) that make up the Internet. Unlike RIP/OSPF, BGP does not select paths using hop count or link cost; instead it uses a rich set of **path attributes** (AS-PATH, NEXT-HOP, LOCAL-PREF, MED, weight, etc.) and a well-defined best-path selection algorithm. A session between routers in **different** ASes is called **eBGP**; a session between routers in the **same** AS is called **iBGP**. BGP is a TCP-based protocol (port 179) and, being designed for policy-based routing at Internet scale, is deliberately conservative – updates are incremental/triggered rather than periodic, and it does not auto-discover neighbors.

4.2 Topology

Two autonomous systems, each with its own internal LAN, connected by a single eBGP link.



IP Addressing Table

Device	AS	Interface	IP Address / Mask
PC0	65001	FastEthernet0	192.168.1.10 /24
R1	65001	Gig0/1 (LAN)	192.168.1.1 /24
R1	65001	Gig0/0 (to R2)	192.0.2.1 /30
R2	65002	Gig0/0 (to R1)	192.0.2.2 /30
R2	65002	Gig0/1 (LAN)	192.168.2.1 /24
PC1	65002	FastEthernet0	192.168.2.10 /24

Procedure

Step 1 – Address all interfaces per the table above and configure the PC gateways.

Step 2 – Configure eBGP on R1 (AS 65001).

```

R1(config)# router bgp 65001
R1(config-router)# neighbor 192.0.2.2 remote-as 65002
R1(config-router)# network 192.168.1.0 mask 255.255.255.0
  
```

Step 3 – Configure eBGP on R2 (AS 65002).

```
R2(config)# router bgp 65002
R2(config-router)# neighbor 192.0.2.1 remote-as 65001
R2(config-router)# network 192.168.2.0 mask 255.255.255.0
```

Step 4 – (Optional) Disable synchronization / set router-id explicitly if using an older IOS image where these are not already sensible defaults:

```
R1(config-router)# bgp router-id 1.1.1.1
R1(config-router)# no synchronization
```

Note

The **network** statement under **router bgp** does **not** enable BGP on an interface the way it does for RIP/OSPF. It simply tells BGP "originate/advertise this prefix if it already exists in my IP routing table" (typically as a connected or static route). This is a key conceptual difference from IGP's and a common source of confusion.

4.3 Verification

```
R1# show ip bgp summary
R1# show ip bgp
R1# show ip route bgp
PC0> ping 192.168.2.10
PC0> tracert 192.168.2.10
```

show ip bgp summary should show the neighbor's BGP state as an uptime value (e.g. 00:02:14) rather than a state name once the session is fully established. **show ip route bgp** should list the remote AS's network with code B.

4.4 Review Questions

1. Why is BGP called a *path-vector* protocol rather than a distance-vector or link-state protocol?
2. What is the difference between eBGP and iBGP, and why does iBGP have a higher default administrative distance (200) than eBGP (20)?
3. In this lab, what would happen to reachability if R1's **network** statement used a prefix that did not exist in its own routing table?

Lab Exercise

Add a third AS (65003) attached to R2 with its own router R3 and LAN, so that R2 now peers with both R1 (AS 65001) and R3 (AS 65003). Verify with **show ip bgp** that PC0 in AS 65001 can reach the LAN in AS 65003, and inspect the **AS-PATH** attribute shown in the BGP table to see how R1 learns about the AS 65003 prefix via AS 65002.

Appendix: Quick Command Reference

Command	Purpose
<code>show ip route</code>	Display the full IP routing table (codes: C=connected, S=static, R=RIP, O=OSPF, B=BGP)
<code>show ip interface brief</code>	Quick summary of interface IPs and up/down status
<code>show running-config</code>	Display the active configuration
<code>ip route <i>dest mask next-hop</i></code>	Add a static route
<code>router rip / version 2 /</code>	Enable RIPv2 and advertise a network
<code>network <i>net</i></code>	
<code>router ospf <i>pid</i> / network <i>net</i></code>	Enable OSPF and advertise a network into an area
<code><i>wildcard</i> area <i>n</i></code>	
<code>router bgp <i>asn</i> / neighbor <i>ip</i></code>	Enable BGP, peer with a neighbor AS, and originate a prefix
<code>remote-as <i>asn</i> / network <i>net</i></code>	
<code>mask <i>mask</i></code>	
<code>show ip protocols</code>	Summarize all active routing protocols and their timers
<code>show ip ospf neighbor</code>	List OSPF neighbor adjacency states
<code>show ip bgp summary</code>	List BGP neighbors and session state/uptime
<code>ping <i>ip</i> / traceroute <i>ip</i></code>	Test reachability / trace the path (Packet Tracer PC syntax)
<code>copy running-config</code>	Save the configuration to NVRAM
<code>startup-config</code>	

End of Lab Manual – Static and Dynamic Routing with Cisco Packet Tracer